Numerical Computing HW#5

1) Provide a proof for: If points $x_0, x_1, \ldots x_n$ are distinct then for arbitrary real values $y_0, y_1, \ldots y_n$, there is a unique polynomial, $P_n(x)$ of degree (at most) n S.T. $P_n(x_i) = y_i$, $0 \le i \le n$

- Assume 2 Interpolating Polynomials $P(x)$ and $Q(x)$, both degree n such that for $0 \le i \le n$, $P(x_i) = y_i$ and $Q(x_i) = y_i$

- A 3rd polynomial $R(x) = P(x) - Q(x)$, implies that for all x_i, $0 \le i \le n$ then $R(x_i) = 0$
  ↳ Implies that $R(x)$ has $n+1$ roots, so x_0 to x_n. Since $R(x)$ has degree n, $R(x)$ can have at most n roots, this is per "the Fundamental Thm of Algebra".
- This implies that $R(x)$ is the zero polynomial thus $P(x) = Q(x)$ and is unique for the given set of points

2) Use Lagrange to find Interpolating Polynomial for table below

X	0	2	3	4
Y	7	11	28	63

Form $P_3(x) = \sum_{i=0}^{3} l_i(x) f(x_i)$ } from Kincaide page 155

$$l_i(x) = \prod_{\substack{j \ne i \\ j=0}}^{n} \left(\frac{x - x_j}{x_i - x_j} \right)$$

- Without l_i,
  $P_n(x) = 7 \cdot l_0(x) + 11 l_1(x) + 28 l_2(x) + 63 l_3(x)$
- Finding l_i's

$$l_0 = \left(\frac{x-x_1}{x_0-x_1}\right)\left(\frac{x-x_2}{x_0-x_2}\right)\left(\frac{x-x_3}{x_0-x_3}\right) = \left(\frac{x-2}{0-2}\right)\left(\frac{x-3}{0-3}\right)\left(\frac{x-4}{0-4}\right) = -24(x-2)(x-3)(x-4)$$

$$l_1 = \left(\frac{x-x_0}{x_1-x_0}\right)\left(\frac{x-x_2}{x_1-x_2}\right)\left(\frac{x-x_3}{x_1-x_3}\right) = \left(\frac{x-0}{2-0}\right)\left(\frac{x-3}{2-3}\right)\left(\frac{x-4}{2-4}\right) = 4(x)(x-3)(x-4)$$

$$l_2 = \left(\frac{x-x_0}{x_2-x_0}\right)\left(\frac{x-x_1}{x_2-x_1}\right),\left(\frac{x-x_3}{x_2-x_3}\right) = \left(\frac{x-0}{3-0}\right)\left(\frac{x-2}{3-2}\right)\left(\frac{x-4}{3-4}\right) = -3(x)(x-2)(x-4)$$

$$l_3 = \left(\frac{x-x_0}{x_3-x_0}\right)\left(\frac{x-x_1}{x_3-x_1}\right)\left(\frac{x-x_2}{x_3-x_2}\right) = \left(\frac{x-0}{4-0}\right)\left(\frac{x-2}{4-2}\right)\left(\frac{x-3}{4-3}\right) = 8(x)(x-2)(x-3)$$

$$L_3(x) = -7 \cdot 24(x-2)(x-3)(x-4) + 44(x)(x-3)(x-4) - 3 \cdot 28(x)(x-2)(x-4) + 8 \cdot 63 (x)(x-2)(x-3)$$

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3) Newtons Interpolation Form

Burden #17 → Divided Difference : $P_n(x) = f[x_0] + \sum_{k=1}^{n} f[x_0, x_1, \dots x_k](x-x_0)\dots(x-x_{k-1})$

	x	$f(x)$	$f[,]$	$f[,,]$	$f[,,,]$
0	0	7	$\frac{11-7}{2-0} = 2$	$\frac{17-2}{3-0} = 5$	
1	2	11	$\frac{28-11}{3-2} = 17$	$\frac{35-17}{3-1} = 9$	$\frac{9-5}{4} = 1$
2	3	28	$\frac{63-28}{4-3} = 35$		
3	4	63			

$a_0 = 7$
$a_1 = 2$
$a_2 = 5$
$a_3 = 1$

$= f(x_0) + a_1(x-x_0) + a_2(x-x_0)(x-x_1) + a_3(x-x_0)(x-x_1)(x-x_2)$

$P_3(x) = 7 + 2(x-7) + 5(x-7)(x-11) + (x-7)(x-11)(x-63)$

* Polynomial Comparison

$N_3(x) = 7 + 2(x-7) + 5(x-7)(x-11) + (x-7)(x-11)(x-63)$

$L_3(x) = -168(x-2)(x-3)(x-4) + 44(x)(x-3)(x-4) - 84(x)(x-2)(x-4) + 504(x)(x-2)(x-3)$

- The Thm says that for all (x_i, y_i) where $0 \le i \le n$, that the polynomial $P_n(x_i) = y_i$.
- If the polynomials are done correctly then $N_3(x_i) = L_3(x_i) = y_i$ implying that the values are the same at the Nodes. The Theorm never mentions intermediate points, so as long as $N_3(x_i) = L_3(x_i)$ the Uniqueness Thm is true

4) A unique polynomial, degree ≤ 2, ST $p(0)=0, p(1)=1, p'(d)=2$ for $d \in [0,1]$ excep the value d, called d_0. Determine d_0 and construct the polynomial for $d \ne d_0$

- $p(x)$ has form $P(x) = ax^2 + bx + c$
- if $p(0) = 0$, $c = 0$ so $\underline{p(x) = ax^2 + bx}$
- if $p(1) = a(1)^2 + b(1) = 1 \Rightarrow \underline{a+b=1}$
- $d_0 = \frac{1}{2}$, Solve for a and b using $2ad + b = 2$ and $a+b=1$,

$a = \frac{1}{2d-1}$ $\quad b = 1 - \frac{1}{2d-1}$

$$\boxed{p(x) = \left(\frac{1}{2d-1}\right)x^2 + \left(1 - \frac{1}{2d-1}\right)x}$$

Find $p'(x)$

$p(x) = ax^2 + bx$

$p'(x) = 2ax + b$

- If $p'(d) = 2$,

$\underline{2 = 2ad + b}$ solve system for d

$\begin{cases} a+b=1 \Rightarrow b = -a \\ 2ad+b=2 \end{cases}$

$\Rightarrow 2ad - a = 2, a(2d-1) = 1,$

$2d-1 \ne 0$

$\boxed{d \ne \frac{1}{2}}$

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5) If g interpolates a function @ $x_0, x_1, \dots x_{n-1}$ and
h interpolates @ $x_0, x_1, \dots x_n$

Then $g(x) + \frac{x_0 - x}{x_n - x_0}[g(x)-h(x)]$ interpolates f @ $x_0, x_1, \dots x_n$

• Only need to check $x_0, x_1 \dots x_n$

• $I(x) = g(x) + \frac{x_0 - x}{x_n - x_0}[g(x)-h(x)]$

• Check x_0

$I(x_0) = g(x_0) + \frac{\cancel{x_0 - x_0}^{\,0}}{x_n - x_0}[g(x)-h(x)] \Rightarrow \underline{I(x_0) = g(x_0) = f(x_0)}$

• Check x_n

$I(x_n) = g(x_n) + \frac{\cancel{x_0 - x_n}^{\,-1}}{\cancel{x_n - x_0}}[g(x_n)-h(x_n)] \Rightarrow I(x_n) = \cancel{g(x_n)} - \cancel{g(x_n)} + h(x_n)$

so $I(x_n) = h(x_n) = f(x_n)$

• If $I(x_0) = f(x_0)$ and $I(x_n) = f(x_n)$ the polynomial $I_n(x)$ interpolates the function $f(x)$. Lagrange and Newton Interpolation guarantee that the intermediate points exist

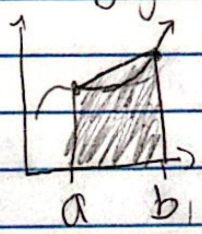
6) Show that the trapezoid rule for Integration results from approximating f by 1st Degree Spline, then using $\int_a^b f(x) \approx \int_a^b S(x)$

Kincaid 203 • Trapezoid Rule: $\int_{x_i}^{x_{i+1}} f(x)\,dx \approx \frac{1}{2}(x_{i+1} - x_i)[f(x_i) + f(x_{i+1})]$

• 1st Degree spline of f(x) on [a,b]

• Spline given by point slope form

Pictoral representation



gives a trapezoid since 1st degree results in a line

$S(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x-a)$

• Taking the integral of S(x): $\int_a^b S(x)$

$= \int_0^b \left(f(a) + \frac{f(b)-f(a)}{b-a}(x-a)\right)dx$

$= \underline{\int_0^b f(a)dx} + \underline{\int_a^b \frac{f(b)-f(a)}{b-a}(x-a)dx}$

$\int_0^b f(a)dx = f(a)(b-a)$

$\int_a^b \frac{f(b)-f(a)}{b-a}(x-a)dx$ $\quad u = x-a$, $du = dx$

$= \frac{f(b)-f(a)}{b-a}\int_0^{b-a} u\,du$

$= \frac{f(b)-f(a)}{b-a} \cdot \frac{(b-a)^2}{2}\Big|_0^{b-a}$

$= \frac{f(b)-f(a)}{b-a}(b-a)$

$= f(a)(b-a) + \frac{f(b)-f(a)}{\cancel{b-a}}(b-a)$

$\boxed{= \left(\frac{f(a)+f(b)}{2}\right)(b-a)} = \int S(x)dx$

Trapezoid rule in a,b terms $\Rightarrow \frac{1}{2}(b-a)(f(b)+f(a))$ ∴ They are the same